PROJECT REPORT

Health360

Healthcare Risk & Lifestyle Analysis

An interactive healthcare data-visualization project using Tableau dashboards, a guided data story, and a responsive companion website.

Project detail	Information
Project name	Health360: Healthcare Risk & Lifestyle Analysis
Prepared by	Aadarsh Barkiya
Project type	Academic data analytics / visualization project
Dataset representation	5,000 sample patient records
Tools	Tableau Public, HTML, CSS, JavaScript
Report date	31 October 2026

Scope note: The 5,000 records represent the supplied sample dataset, not verified real patients. This project is educational and is not a diagnostic tool.

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Medical disclaimer: Health360 does not provide medical advice and should not be used to make individual healthcare decisions.

1. INTRODUCTION

1.1 Project Overview

Health360 is a student project that explores patterns in a 5,000-record sample healthcare dataset. It combines Tableau visualizations with a responsive HTML website so viewers can examine patient age groups, lifestyle indicators, BMI categories, blood pressure and recorded heart-disease outcomes. A companion Tableau story provides a guided narrative alongside the interactive dashboard.

1.2 Purpose and Objectives

- Present sample healthcare data through clear, interactive visualizations.
- Compare records across six age groups: Adult, Middle Age, Old, Senior, Teenager and Young Adult.
- Explore exercise levels, fatigue, alcohol consumption and smoking patterns.
- Compare average blood pressure across age groups and examine BMI categories alongside recorded heart-disease outcomes.
- Provide one website interface from which viewers can switch between the Tableau dashboard and story.

2. IDEATION PHASE

2.1 Problem Statement

Healthcare datasets can contain many variables that are difficult to interpret as raw rows and columns. Health360 addresses this presentation challenge by organizing selected demographic, lifestyle and health indicators into charts and an accompanying story. Its focus is exploratory visualization rather than prediction or diagnosis.

2.2 Design Decisions

- Use age-group comparisons to make demographic patterns easier to inspect.
- Use categorical charts and a heatmap to explore lifestyle-related variables.
- Present BMI and heart-disease measures together for visual comparison.
- Keep the dashboard and guided story accessible from the same website.
- Clearly communicate that the data is sample data, not validated clinical evidence.

2.3 Selected Visualizations

Visualization	Purpose
Patient distribution by age group	Compare records across six age segments.
Exercise-level distribution	Show the distribution of exercise categories.
Average blood pressure by age group	Compare average blood-pressure values across age segments.
BMI and heart-disease views	Explore recorded outcomes across BMI categories.
Lifestyle-risk heatmap	Compare fatigue patterns across age groups and exercise levels.
Alcohol and smoking by age group	Inspect lifestyle indicators across demographic groups.
Heart-disease risk by age group	Compare recorded patterns between age segments.

3. REQUIREMENT ANALYSIS

3.1 User Journey

Stage	User action	Expected experience
Discover	Open the Health360 website or Tableau link.	Understand project scope and available views.
Explore	Choose Dashboard or Data Story.	View the selected Tableau visualization.
Interpret	Inspect charts and compare categories.	Identify descriptive patterns in sample data.
Share	Open Tableau Public links or share website files.	Review project outputs.

3.2 Functional Requirements

- Responsive website introducing the Health360 project.
- Two embedded Tableau views: Healthcare Risk Analysis dashboard and Patient Risk Analysis story.
- Tabs switch between dashboard and story without leaving the website.
- Direct links open each visualization on Tableau Public.
- Project overview, exploration themes and healthcare-use disclaimer.

3.3 Conceptual Data Flow

Input	Analysis	Output
Sample healthcare dataset	Group and summarize selected fields in Tableau.	Charts and summary views.
Age, BMI and lifestyle fields	Compare categories and measures across groups.	Demographic, lifestyle and health visualizations.
Published Tableau views	Embed using Tableau Embedding API v3.	Dashboard and story in one website.

This is a high-level workflow description. It does not claim a separate database, live hospital integration or an undocumented backend.

3.4 Technology Stack

Technology	Role
Tableau Public	Create and publish the interactive dashboard and guided story.
HTML	Structure website content and visualization area.
CSS	Responsive layout, typography, colors and styling.
JavaScript	Switch embedded views and update titles, descriptions and links.
Tableau Embedding API v3	Load Tableau visualizations inside the website.

4. PROJECT DESIGN

4.1 Problem-Solution Fit

The project makes a multi-variable sample dataset easier to explore through visual comparisons. The dashboard supports direct chart exploration, while the story organizes the analysis into a guided sequence. The companion website provides a common entry point for both views.

4.2 Proposed Solution

- A responsive Health360 website with project introduction and dataset overview.
- A visualization workspace with Dashboard and Data Story tabs.
- Embedded Tableau views with links to open them directly on Tableau Public.
- Sections describing demographic, lifestyle and health-indicator themes.
- A disclaimer explaining the educational scope and limitations.

4.3 Solution Architecture

Presentation layer	Visualization layer	Access layer
HTML/CSS/JavaScript website	Tableau dashboard and story	Local website files and Tableau Public links

The website loads the selected Tableau view from its published URL. JavaScript changes the embedded visualization source and updates the corresponding title, description and external link.

5. PROJECT PLANNING & SCHEDULING

Phase	Work performed	Output
Define	Identify project theme and healthcare indicators.	Project scope.
Prepare data	Work with supplied sample healthcare dataset.	Dataset ready for visualization.
Visualize	Build demographic, lifestyle and health worksheets, dashboard and story.	Published Tableau views.
Develop website	Create responsive website and connect both Tableau views.	HTML/CSS/JavaScript website.
Review	Check Tableau links and website tab switching.	Demonstration-ready project.

A dated task log and individual team assignments were not supplied, so no completion dates or roles are invented.

6. FUNCTIONAL AND PERFORMANCE TESTING

Test area	Check	Expected result
Website launch	Open index.html in a modern browser.	Website structure and styling display.
Dashboard tab	Select 01 Dashboard.	Healthcare Risk Analysis dashboard loads.
Story tab	Select 02 Data Story.	Patient Risk Analysis story loads.
Tab state	Switch between both tabs.	Active tab and view details update.
Tableau link	Use Open on Tableau.	Corresponding Tableau view opens.
Responsive layout	View on desktop and narrow screens.	Content remains readable.

These are functional checks and expected outcomes, not measured performance benchmarks. Load time depends on browser, network and Tableau Public availability.

7. RESULTS

7.1 Tableau Dashboard

The published Health360 Healthcare Risk Analysis dashboard brings together selected demographic, lifestyle and health measures. The project visualizations include patient distribution by age group, exercise-level distribution, average blood pressure by age group, BMI-category comparisons with heart-disease outcomes, fatigue patterns, alcohol and smoking patterns, and heart-disease patterns by age group.

7.2 Tableau Story

The companion Patient Risk Analysis story provides a guided way to move through the project findings. It complements the dashboard by presenting the analysis in a narrative sequence rather than requiring viewers to explore every chart independently.

7.3 Website

The website combines both Tableau resources in one responsive page. Dashboard and Data Story tabs load the selected visualization in the same embedded viewing area. Each view can also be opened directly on Tableau Public.

7.4 Published Outputs

Tableau Dashboard: Open Health360 Healthcare Risk Analysis

Tableau Story: Open Health360 Patient Risk Analysis

Website: Supplied as downloadable HTML/ZIP files. A public deployment URL was not provided, so this report does not claim it is live online.

8. ADVANTAGES & DISADVANTAGES

8.1 Advantages

- Combines multiple healthcare indicators in an organized visual presentation.
- Supports comparisons across age groups and lifestyle categories.
- Provides both an exploratory dashboard and a guided story.
- Makes both Tableau views accessible through one website interface.
- Uses Tableau Public links for online demonstration.

8.2 Limitations

- The dataset is sample/illustrative and is not verified clinical patient evidence.
- Visual patterns do not establish causation.
- The project is not a diagnostic, predictive or treatment-support system.
- Embedded Tableau views require an internet connection and Tableau Public availability.
- No live hospital-record integration or clinical validation is included.

9. CONCLUSION

Health360 demonstrates how Tableau and a responsive web interface can make sample healthcare data easier to explore. The dashboard presents demographic, lifestyle and health-indicator comparisons, while the companion story provides a guided narrative. The website brings both Tableau views together through a simple tab-based interface. The project is an educational data-visualization prototype, not medical evidence or a clinical decision tool.

10. FUTURE SCOPE

- Document dataset source, provenance, definitions and data-cleaning steps.
- Add further visual comparisons where supported by the dataset.
- Improve accessibility with keyboard testing and descriptive chart labels.
- Deploy the website publicly and add a verified website URL.
- Conduct formal usability and browser/device testing.
- Apply privacy safeguards and clinical validation before any real-world healthcare application.

11. APPENDIX

11.1 Project Links

Resource	Link / status
Tableau Dashboard	https://public.tableau.com/views/Health360HealthcareRiskAnalysis_17932895178070/Health360HealthcareRiskAnalysis
Tableau Story	https://public.tableau.com/views/Health360PatientRiskAnalysis/Health360PatientRiskAnalysis
Website source	Provided separately as index.html and ZIP; no public deployment URL supplied.
GitHub repository	Not supplied for this report.
Demo video	Not supplied for this report.
Dataset URL	Not supplied; dataset described as sample data.

11.2 Source Code and Implementation

The companion website uses HTML, CSS and JavaScript. It uses Tableau Embedding API v3 to load the published dashboard and story. The tab controls update the embedded Tableau source and corresponding view information.

11.3 Dataset Note

Project materials describe a 5,000-record sample healthcare dataset. A separate public dataset URL and full data dictionary were not supplied. The 5,000 figure refers to the sample dataset represented in the project, not verified real patients.

11.4 Disclaimer

Health360 is for academic and educational data exploration only. It is not intended to diagnose, predict, prevent or treat any medical condition. Consult qualified healthcare professionals for medical advice.

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